

Revision Question Alternative Set 2

Question 1

- a) Write a query to rank products within each category based on their price.
- Use the RANK() or DENSE_RANK() window function.
 - Partition by category and order by price descending.
 - Output: category, product_name, price, and price_rank
- b) Write a query to find all customers who have only made transactions where the total_amount was less than \$10.
- Use a subquery with NOT IN or EXCEPT to exclude customers who have made any transaction $\geq$ \$10.
 - Output: first_name and last_name
- c) Write a query to list every transaction, the name of the customer, and the total number of unique items in that transaction.
- Join customers, transactions, and transaction_items.
 - Use COUNT(DISTINCT product_id) and group the results.
 - Output: transaction_id, first_name, last_name, and unique_item_count

d) Categorize transactions based on how recently they were made compared to a specific date (e.g., '2024-10-05').

- Use a CASE statement to label transactions as:
- 'Recent' if the transaction_date is on or after '2024-10-05'.
- 'Old' if the transaction_date is before '2024-10-05'.
- Output: transaction_id, transaction_date, and recency_status

e) For each customer, calculate the difference in total_amount between their current transaction and their previous one.

- Use the LAG() window function partitioned by customer_id and ordered by transaction_date.
- Output: customer_id, transaction_date, total_amount, and difference_from_previous.

f) Calculate what percentage of the total stock quantity each product represents within its own category.

- Use the SUM() window function partitioned by category to get a "Category Total."
- Divide the individual stock_quantity by that "Category Total."
- Output: product_name, category, and percentage_of_category_stock.